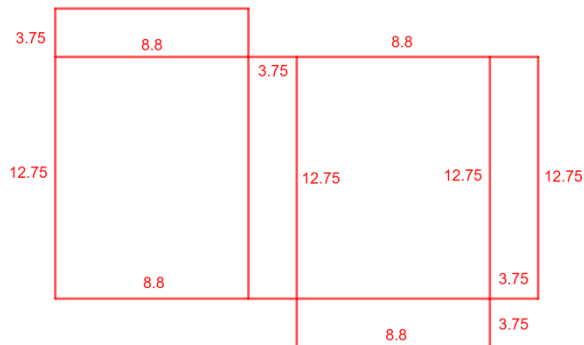


Grade 6 Accelerated
Unit 8
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

A box of cereal measures 8.8 inches (in.) long, 3.75 in. wide, and 12.75 in. tall.

1. Sketch the net of the cereal box.



2. Label the dimensions of each face on the net.

3. Find the area of each face. Include units.

Front: $8.8 \times 12.75 = 112.2 \text{ in}^2$

Side: $12.75 \times 3.75 = 47.8125 \text{ in}^2$

Back: $8.8 \times 12.75 = 112.2 \text{ in}^2$

Top: $8.8 \times 3.75 = 33 \text{ in}^2$

Side: $12.75 \times 3.75 = 47.8125 \text{ in}^2$

Bottom: $8.8 \times 3.75 = 33 \text{ in}^2$

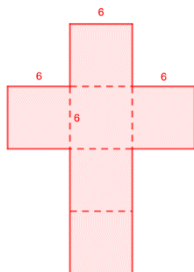
4. Add the area of the faces to find the surface area of the cereal box. Include units.

$$2(112.2) + 2(47.8125) + 2(33) = 224.4 + 95.625 + 66 \\ = 386.025 \text{ in}^2$$

Sharee is designing boxes for the rings she sells. The ring box has measurements 6 centimeters (cm) long, 6 cm wide, and 6 cm high.

5. Sketch and label the net of the ring box.

All sides are 6 cm.



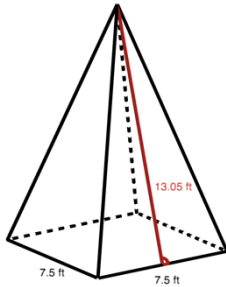
6. What is the surface area of the ring box? Include units.

$$\begin{aligned}\text{Front/back: } 2(6 \times 6) &= 72 \text{ cm}^2 \\ \text{Side/Side: } 2(6 \times 6) &= 72 \text{ cm}^2 \\ \text{Top/Bottom: } 2(6 \times 6) &= 72 \text{ cm}^2\end{aligned}$$

$$72.3 = 216 \text{ cm}^2$$

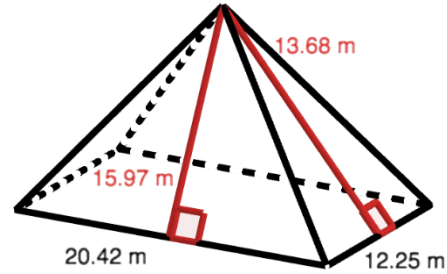
For questions 7–8, find the surface area each pyramid. Include units.

7.



$$\begin{aligned}\square: 7.5 \times 7.5 &= 56.25 \text{ ft}^2 \\ \triangle: \frac{1}{2} \times 7.5 \times 13.05 &= 48.9375 \text{ ft}^2 \\ 56.25 + 4(48.9375) & \\ &= 252 \text{ ft}^2\end{aligned}$$

8.



$$\begin{aligned}\square: 20.42 \times 12.25 &= 250.145 \text{ m}^2 \\ \triangle_1: \frac{1}{2} \times 20.42 \times 15.97 &= 163.0537 \text{ m}^2 \\ \triangle_2: \frac{1}{2} \times 12.25 \times 13.68 &= 83.79 \text{ m}^2 \\ 250.145 + 2(163.0537) + 2(83.79) & \\ &= 743.8324 \text{ m}^2\end{aligned}$$

9. Calculate the surface area of a rectangular pyramid that has a square base with a side length of 14.5 centimeters (cm), which is the base of the triangle with a slant height of 14.2 cm.

$$\begin{aligned}14.5 \times 14.5 &= 210.25 \text{ cm}^2 \\ \frac{1}{2} \times 14.5 \times 14.2 &= 102.95 \text{ cm}^2 \\ 210.25 + 4(102.95) &= 622.05 \text{ cm}^2\end{aligned}$$

10. Calculate the surface area of a rectangular pyramid that has a rectangular base with a side length of 12 yards (yd), which is the base of a triangle with a slant height of 11.66 yd, and a side length of 15 yd, which is the base of a triangle with a slant height of 12.5 yd.

$$\begin{aligned}12 \times 15 &= 180 \text{ yd}^2 \\ \frac{1}{2} \times 12 \times 11.66 &= 69.96 \text{ yd}^2 \\ \frac{1}{2} \times 15 \times 12.5 &= 93.75 \text{ yd}^2 \\ 180 + 2(69.96) + 2(93.75) &= 507.42 \text{ yd}^2\end{aligned}$$